Ghazi University

Assigment

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Class: BSCS 'E'

Subject: PF (Lab)

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CamScanner

BS CS (E)

Lab-1

Task 1:

Run the following program and check the output

```
#include<iostream>
using namespace std;
int main()
{
    cout<<"hello world"<<endl;
    System ("pause");
    return 0;
}
```

Output:-

hello world

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Task: 2

Write a C++ program to print the following lines
You are 10 years old.
You are too young to play the game.

```
#include<iostream>
  using namespace std;
int main ()
{
  Cout<<"you are 10 years old."<<endL;
  Cout<<"you are too young to play the game."<<endL;
  return 0;
}
```

output:- You are 10 years old
You are too young to play the game.

Task :- 3

```
#include<iostream>
  using namespace std;
int main ()
{
  Cout<<"*"<<endL;
  cout<<"**"<<endL;
  Cout<<"***"<<endL;
  Cout<<"****"<<endl;
  return 0;
}
```

Output:-

*
**
***
****

BS CS (E)

Lab-02

Task: 1

```
#include<iostream>
using namespace std;
 int main()
{ Char Letter;
   Letter = 'A';
cout << letter<< '\n';
  Letter = 'B';
 cout << letter<< '\n';
 cin >> Letter;
 return 0;
 }
```

```
#include<iostream>
using namespace std;
 int main ()
{ Char Letter;
 Letter=65;
cout << Letter<< '\n';
Letter = 66;
cout << Letter<< '\n';
cin>> Letter;
return 0;
}
```

```
#include<iostream>
using namespace std;
int main()
{ int letter
Letter='A';
Cout<<Letter<< '\n';
Letter = 'B';
Coat<< Letter<< '\n';
cin >> Letter;
 return 0;
}
```

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Task :- 2

```
# include <iostream>
using namespace std;
int main ()
{
  int num1, num2;
  cout<<"Enter first integer:";
  cin>> num1;
  cout<<" Enter second integer:";
  cin >> num2;
  cout << " Addition (+) :" << num + num2 << endl;
cout<<" substraction (-) :" << num1 - num2 << endl;
cout << " Multiplication (*) : " << num1 * num2 << endl;
if (num2 != 0) {
  cout<<" Division (/) :" <<(double) num1/num2<<endl;
  } else {
  cout<< " Division (/) : Error! Division by zero." <<endl;
  }
  return 0;
}
```

Task :- 3

```
# include "stdafx.h"
# include <iostream>
 using namespace std;
 int main()
 { int val1;
    int val2;
    int val3;
 cout<<" please enter your 3 numbers:";
  cin >> val1 >> val2 >> val3;
cout << val1 << " \n" << val2 << " \n" << val3 << " \n";
cout << val3 << " \n" << val2 << " \n" << val1 << " \n";
  system (" PAUSE");
  return 0;
  }
```

Lab:- 03

Correct the errors

Task:1

i) Write Correct Program

```
#include<iostream>
using namespace std;
 int main()
{
  double number1, number2, sum;
Cout<<"Enter another number:";
 cin >> number1;
Cout<<"Enter another number:";
Cin >> number2;
Sum = number1 + number2;
Cout<<"The sum of the two numbers is"<<
Sum<< endl;
return 0;
}
```

output:

```
Enter a number: 10
Enter a another number: 20
The sum of the two number: 30
```

ii) Corrected Program

```
#include<iostream>
using namespace std;
int main()
{ int number1, number2;
  float quotient;
Cout<<"Enter two number and I
will divide !n";
Cout"The first by the second for you !n";
 cin>> number1>> number2;
 if (number2 != 0)
 { quotient = static_cast<float>(number1)/
  number2;
Cout<<"Quotient="<< quotient<<endl;
 }
else {
cout<<"Error: Division by zero is not
  allowed." endl;
  }
 return 0;
}
```

Task 2

```
#include<iostream>
using namespace std;
int main()
{
  int num1=28;
  int num2=32;
  int num3=37;
  int num4=24;
  int num5=33;
int sum;
double average;
sum=num1+num2+num3+num4+num5;
average=sum/5.0;
cout<<"sum="<<sum<<endl;
cout<<"Average="<<average<<endl;
 return 0;
}
```

Out put:

Sum 154

Average 30.8

Lab :- 04

Task :- 1

Check Positive or Negative number

```
#include<iostream>
using namespace std;
int main()
{ int number;
 cout<<" Enter a number:";
 cin>> number;
 if (number>0)
{cout<<" Numbe is positive";
 }
 else
{
  cout<<" Number is negative";
 }
  return 0;
 }
```

Out Puts:-

```
Enter a number:5
Number is Positive
```

```
#include<iostream>;
using namespace std;
int main()
{ int number;
 cout<<"Enter an integer:";
 cin>> number;
 if (number>0)
{
 cout<<"Number is positive";
}
else if (number<0)
{ cout<<"Number is negative";
 }
  else
{
  cout<<"Number is zero";
  }
 return 0;
  }
```

out puts:-

input	output
8	Number is positive
-4	Number is negative
0	Number is zero

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Task 3:

Calculate Grade using Marks

```
# include <iostream>
using namespace std;
int main()
{
  int mark;
cout<<"Enter your marks:";
cin>> mark;
if (mark>=75)
{
  cout<<" Your grade is A";
 }
 else if (mark>=60)
{
  cout<< "your grade is B";
  }
else if (mark>=45)
{
  cout<< "your grade is C";
  }
  else
  {
   cout<< "your grade is D";
  }
  return 0;
  }
```

Output:

Enter your marks: 82
Your grade is A

Section: BS CS(E)

Subject: Programming Fundamental

Roll No: 39

Submitted:

Lab-05

Task: 1 Point table

```
#include<iostream>
using namespace std;
int main()
{
    int num;
cout<<"Enter any number:";
cin >> num;
for (int i=1; i<=10; i++)
  {
    cout<<num<<"*"<<i<<"="<<num*i<<endl;
    }
    return 0;
    }
```

Task 2:

find Power of any number

```
#include<iostream>
using namespace std;
int main()
{
  int num, Power, result=1;
Cout<<"Enter number:";
 cin>> num;
Cout<<" Enter Power:";
cin>> Power;
for(int i=1; i<=Power; i++)
 {
   result=result * num;
 }
Cout<<"Answer="<<result;
return 0;
}
```

Task 3

Sum of Natural number

```
#includ<iostream>
using namespace std;
int main()
{ int n, sum=0;
Cout<<" Enter a number:";
cin>>n;
for(int i=1; i<=n; i++)
{
  sum= sum +i;
}
 Cout<<"sum="<<sum;
 return 0;
}
```

Task 4:

Find Sum of

```
#include<iostream>
using namespace std;
int main()
{
  int num, digit, sum=0;
Cout<<" Enter a number:";
 cin>> num;
while(num>0)
{ digit = num % 10;
 sum = sum + digit;
} num = num / 10;
Cout<<"sum of digits="<<sum;
return 0;
}
```

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```
int n, sum=0;
Cout<<"Enter number:";
cin>> n;
while(n>0)
{
 sum = sum + n % 10;
  n=n /10;
 }
Cout<<"sum="<<sum;
return 0;
 }
```

...torial of a Number:

```
#include<iostream>
using namespace std;
int main()
{
  int n;
 long long fact=1;
cout<<"Enter a number:";
 cin>>n;
for(int i=1;i<=n;i++)
 {
  fact=fact*i;
 }
  cout<<"Factorial="<<fact;
  return 0;
 }
```

Task: 7

Print the star Pattern

```
#include<iostream>
using namespace std;
int main()
{
  for(int i=1;i<=7;i++)
  {
    for(int j=1;j<=i;j++)
    {
      cout<<"*";
    }
```

```
cout << endl;
}
return 0;
}
```